

Understanding Diabetes Progression

A regression analysis of 442 patients. Final Project, Scenario 1

1. Introduction and Data

This report looks at which patient traits are linked to diabetes progression. The dataset has 442 patients. Each row is one patient. It includes an outcome (a progression score measured one year after baseline) and ten baseline traits: age, sex, body mass index (BMI), average blood pressure (BP), and six blood measures called s1 through s6.

Sex is written as 1 or 2 in the file, but it is a category, not a number you can add up. It was changed to a factor so R treats it as a category. The data has 442 full rows, no missing values, and no repeated rows. No strange values were found, so no rows were removed.

This is real health data, so it should be treated with care. Even though there are no names attached, patients did not choose to have their data used for a class project. Any patterns found here are about the group as a whole, not any single person, and should not be treated as medical advice.

2. Exploring the Data

Progression has a mean of 152.1 and a median of 140.5. It ranges from 25 to 346, with a lot of spread (standard deviation 77.1). Figure 1 shows the shape is uneven: many patients have lower scores and a smaller number have much higher scores.

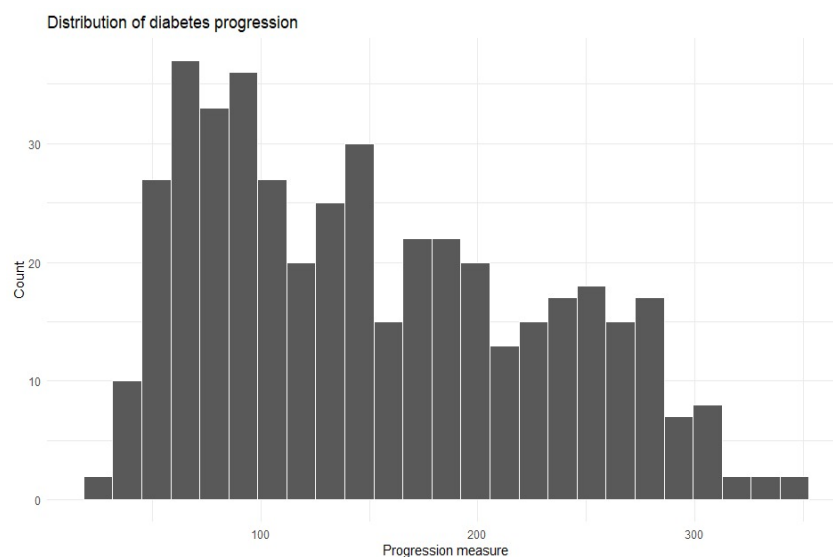


Figure 1. Distribution of diabetes progression.

BMI has a mean of 26.4 and a median of 25.7 (range 18.0 to 42.2). Figure 2 shows a fairly even, bell shaped pattern with a small tail toward higher values.

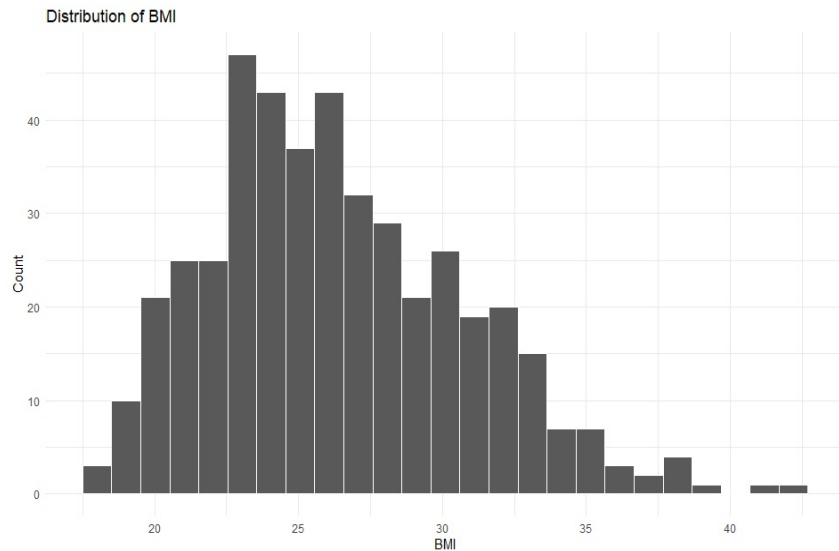


Figure 2. Distribution of BMI.

As one more check, a boxplot of progression (Figure 3) was made to look for outliers and to see the spread more clearly. The box covers the middle half of patients (from 87 to 211.5), and the whiskers reach the lowest and highest values with no points flagged as clear outliers.

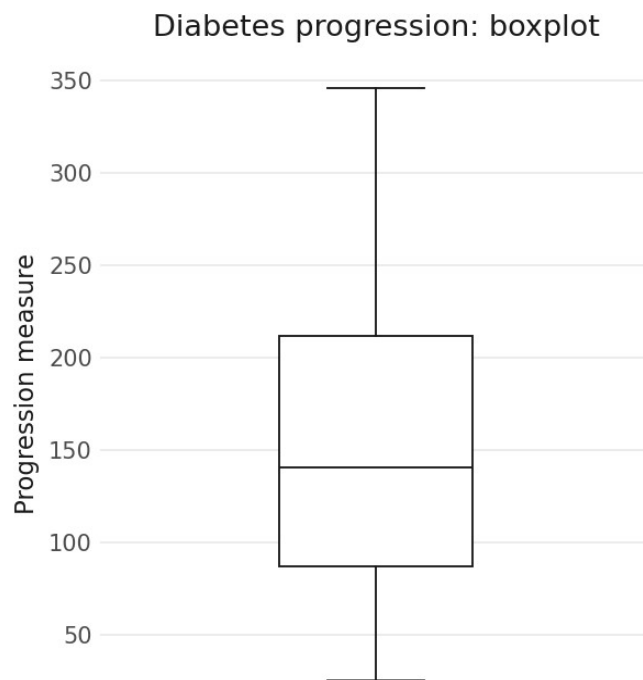


Figure 3. Diabetes progression, shown as a boxplot.

Overall, the data is clean and ready to use. The main thing to remember before modelling is that progression is not evenly spread. That is worth keeping in mind when we check the model later.

3. Looking at Relationships

A correlation matrix shows how strongly the numeric variables move together (sex is left out since it is a category). Figure 4 shows this as a heatmap.

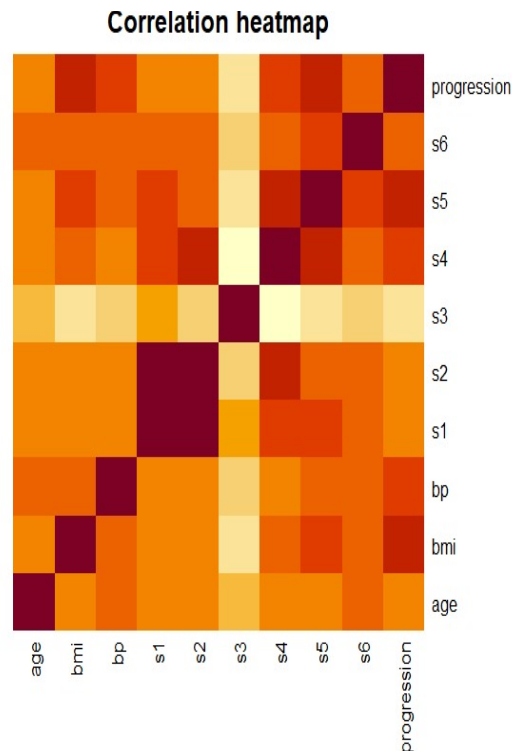


Figure 4. Correlation heatmap of the numeric variables.

BMI ($r = 0.59$) and s5 ($r = 0.57$) have the strongest link to progression. Blood pressure ($r = 0.44$) and s4 ($r = 0.43$) follow. Age, s1, and s2 have a weaker link (all under 0.22).

Some predictors overlap heavily with each other. s1 and s2 move almost together ($r = 0.90$). s3 and s4 also move closely, in opposite directions ($r = -0.74$). This kind of overlap can cause problems if too many related variables are put in one model together, since they carry much of the same information.

Two scatterplots show the strongest patterns with the outcome:

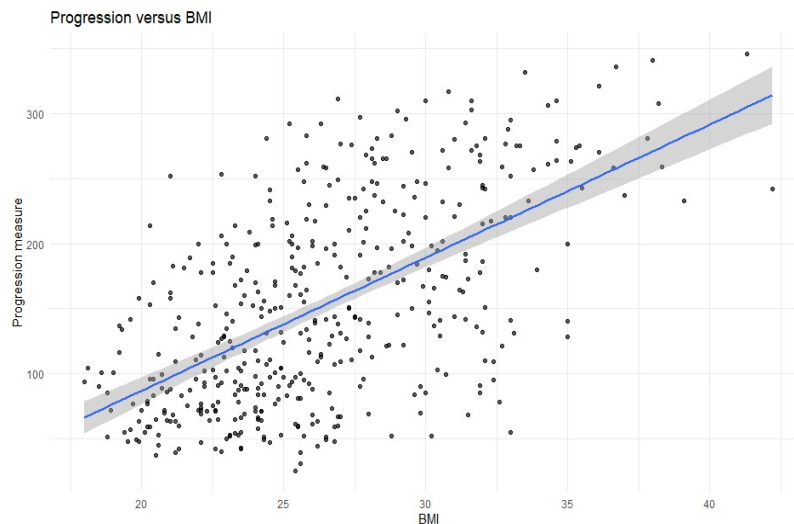


Figure 5. Progression versus BMI.

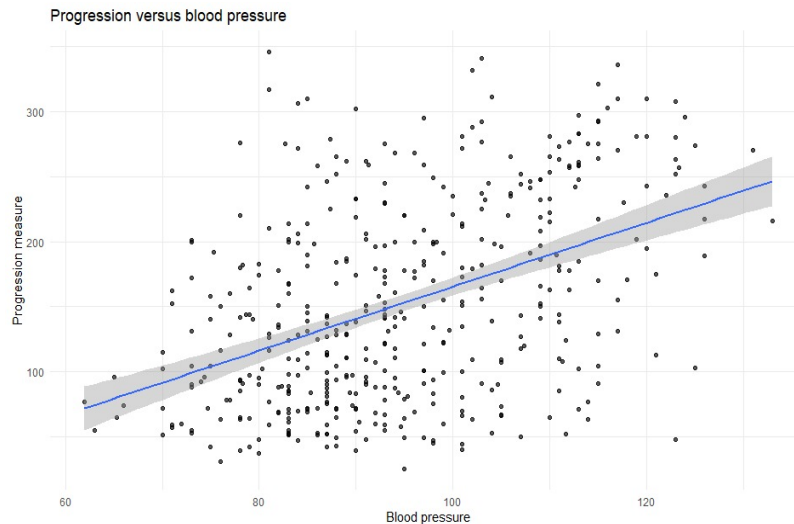


Figure 6. Progression versus blood pressure.

BMI, s5, and blood pressure look like the best choices for a model. Each has a fairly steady, roughly straight line relationship with progression. Unlike the s1/s2 and s3/s4 pairs, they are only loosely linked to each other (all pairwise correlations between 0.39 and 0.45), so combining them should not cause much overlap.

4. Building the Regression Models

A simple model was fit using just BMI, since it had the strongest single link to progression:

$$\text{progression} = -117.77 + 10.23 \times \text{BMI}$$

Each one unit rise in BMI goes with a 10.23 point rise in the progression score, on average. The starting value of -117.8 has no real meaning on its own, since a BMI of 0 never happens. This model explains about 34% of the differences between patients (R squared = 0.344).

A second model adds blood pressure and s5:

$$\text{progression} = -334.88 + 6.50 \times \text{BMI} + 0.90 \times \text{BP} + 49.58 \times \text{s5}$$

Holding BP and s5 steady, each one unit rise in BMI goes with a 6.50 point rise in progression. This is smaller than before because BMI shares some of its link with BP and s5. Holding BMI and s5 steady, each one unit rise in BP goes with a 0.90 point rise. Holding BMI and BP steady, each one unit rise in s5 goes with a 49.58 point rise, the largest jump of the three, though s5 is measured on a much smaller scale so this is not directly comparable in size.

Model	R squared	Adjusted R squared	Residual error
Simple: BMI only	0.344	0.342	62.52
Multiple: BMI + BP + s5	0.480	0.477	55.78

The second model explains more (R squared rises from 0.344 to 0.480) and its errors are smaller on average. This makes sense, since adding useful predictors rarely hurts fit. But a higher R squared is not the only thing that matters. The second model is also more useful because it separates the roles of BMI, blood pressure, and blood chemistry instead of letting BMI take credit for all of them.

5. Checking the Model

Figure 7 plots the leftover errors (residuals) against the model's predicted values. The points spread fairly evenly above and below zero, without a strong curve, though the spread grows a little at higher predicted values. This is not a major problem, but it is a sign the straight line model is a simplification rather than a perfect fit, which matches the uneven shape of progression seen earlier.

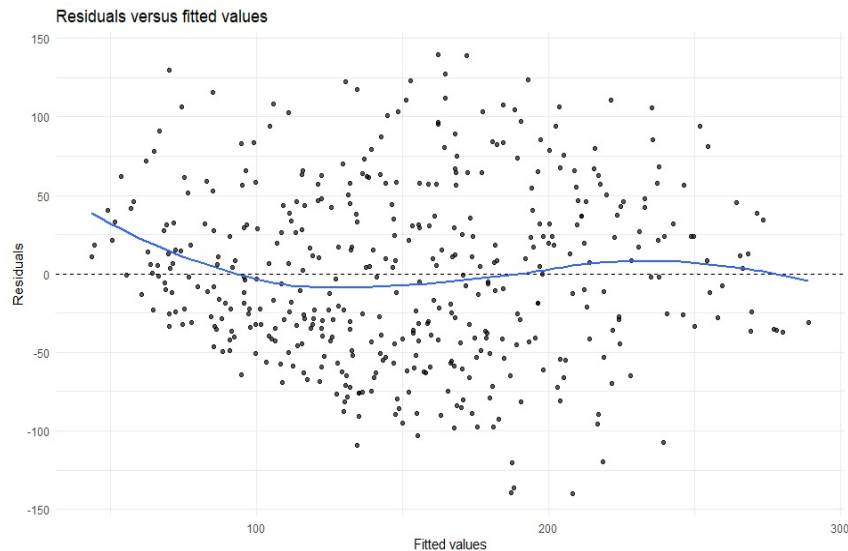


Figure 7. Residuals versus fitted values.

To check for overlap between predictors, variance inflation factors (VIF) were calculated: BMI = 1.35, BP = 1.27, s5 = 1.34. All are well below the usual concern level of 5, so this particular combination of three predictors does not cause a real overlap problem, even though other variables in the dataset (like s1/s2) do overlap heavily and were left out for that reason.

Looking closer at two key results from the second model:

- BMI: 6.50, with a range of 5.13 to 7.87 that we are 95% confident contains the true value. p is below 0.001, so this result is very unlikely to be due to chance.
- s5: 49.58, with a range of 38.00 to 61.15. p is also below 0.001.

All three predictors are statistically significant, meaning they are very unlikely to be random noise in a sample this size. But significant does not mean large or important in practice. With over 440 patients, even small effects can show up as significant. Since the model only explains about half the differences in progression, it is useful for spotting general patterns, but it should not be used to predict any one patient's outcome precisely.

6. What We Can and Cannot Conclude

This dataset is observational. Patients were simply measured as they were, not assigned by chance to different BMI or blood pressure levels the way they would be in an experiment. Because of this, the results show links, not proof of cause and effect. A strong BMI result does not prove that lowering BMI would lower progression. It is just as possible that both are driven by something else.

Other factors that are not in this dataset could affect the results, such as medication, diet, exercise, how long someone has had diabetes, or income and access to care. If any of these affect both a predictor and progression, the coefficients could be misleading. The blood measures were likely taken at one visit, so timing or lab differences could also add noise.

It is also unclear how well these 442 patients represent people with diabetes more broadly, since details about age range, location, or clinical setting are not given. The results may not carry over directly to a different group of patients.

What the analysis does support: BMI, blood pressure, and the s5 blood measure have a clear, reliable link to diabetes progression in this group, and together they explain close to half of the differences seen between patients. This is a useful pattern worth exploring further. What it does not support: that changing these measures would cause progression to slow down, that these three variables alone can predict any one patient's outcome precisely, or that the same pattern would definitely hold in a different group of people.